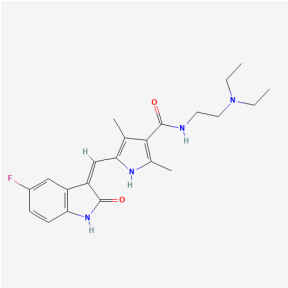
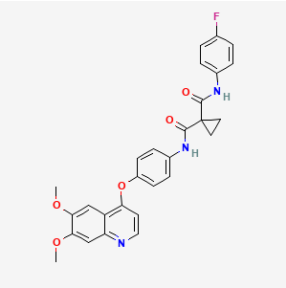
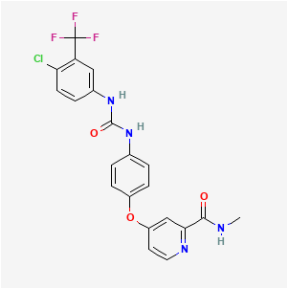


Sunitinib
MF:$C_{22}H_{27}FN_4O_2$
CN: H_1
MV: 324.1

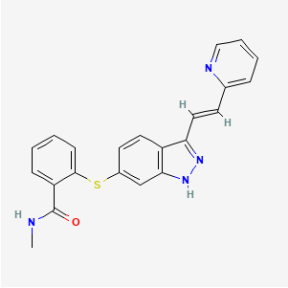
Pazopanib
MF:$C_{21}H_{23}N_7O_2S$
CN: H_2
MV: 310.4



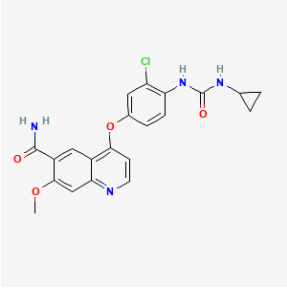
Cabozantinib
MF:$C_{28}H_{24}FN_3O5$
CN: H_3
MV: 359.0



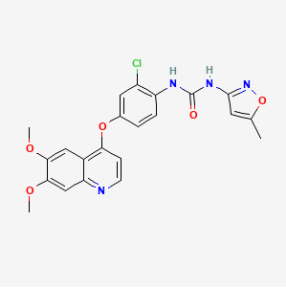
Sorafenib
MF:$C_{21}H_{16}ClF_3N_4O_3$
CN: H_4
MV: 319.5



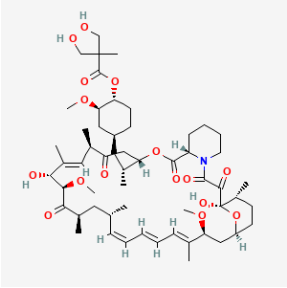
Axitinib
MF:$C_{22}H_{18}N_4OS$
CN: H_5
MV: 284.8



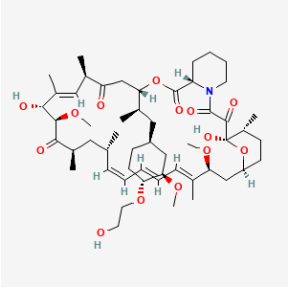
Lenvatinib
MF:$C_{21}H_{19}ClN_4O_4$
CN: H_6
MV: 290.6



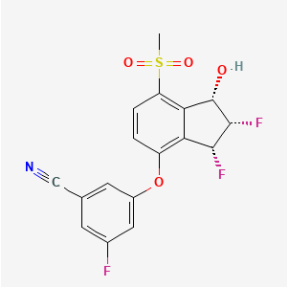
Tivozanib
MF:$C_{22}H_{19}ClN_4O_5$
CN: H_7
MV: 320.0



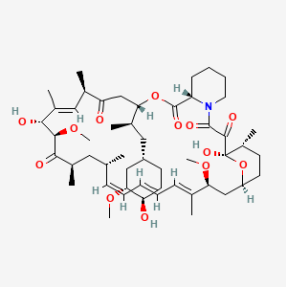
Temsirolimus
MF:$C_{56}H_{87}NO_{16}$
CN: H_8
MV: 853.1



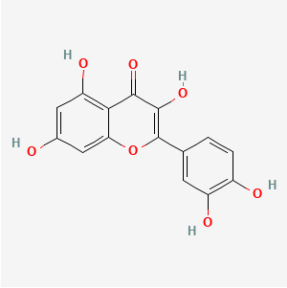
Everolimus
MF:$C_{53}H_{83}NO_{14}$
CN: H_9
MV: 811.2



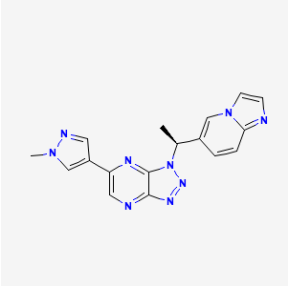
Belzutifan
MF:$C_{17}H_{12}F_3NO_4S$
CN: H_{10}
MV: 244.7



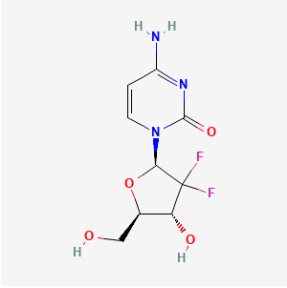
Sirolimus
MF:$C_{51}H_{79}NO13$
CN: H_{11}
MV: 773.5



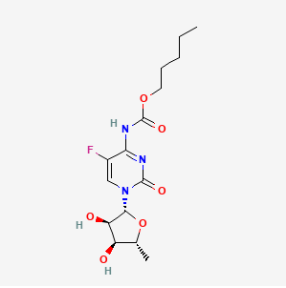
Quercetin
MF:$C_{15}H_{10}O_7$
CN: H_{12}
MV: 168.0



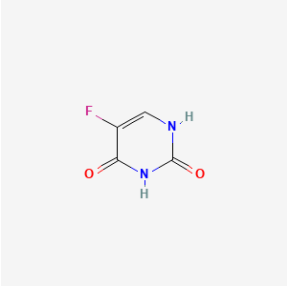
Sirolimus
MF:$C_{17}H_{15}N_9$
CN: H_{13}
MV: 222.5



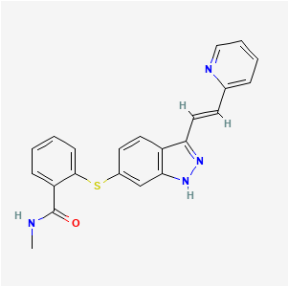
Gemcitabine
MF:$C_9H_{11}F_2N_3O_4$
CN: H_{14}
MV: 142.3



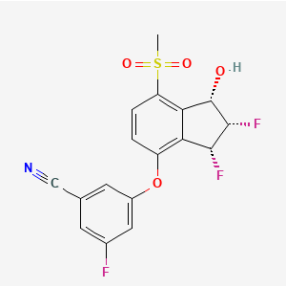
Capecitabine
MF:$C_{15}H_{22}FN_3O6$
CN: H_{15}
MV: 240.5



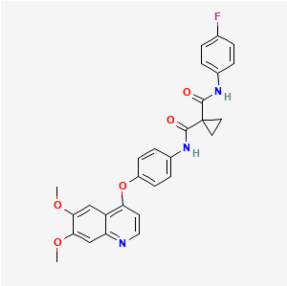
5-Fluorouracil
MF:$C_4H_3FN_2O_2$
CN: H_{16}
MV: 84.6



Inlyta
MF:$C_{22}H_{18}N_4OS$
CN: H_{17}
MV: 284.8

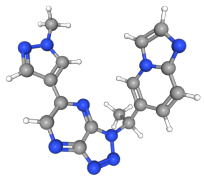


Welireg
MF:$C_{21}H_{24}ClN_7O_2S$
CN: H_{18}
MV: 244.7

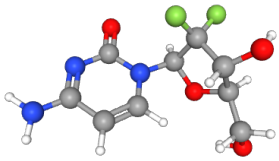


Cabometyx
MF:$C_{28}H_{24}FN_3O5$
CN: H_{19}
MV: 359.0

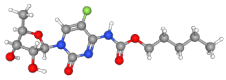
Figure 1: 2D molecular structures of drug molecules



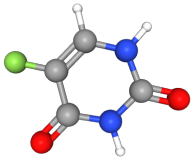
(a) Sunitinib



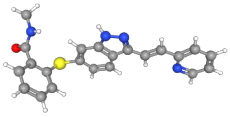
(b) Gemcitabine



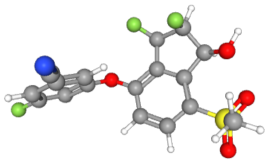
(c) Sorafenib



(d) 5-Fluorouracil



(e) Inlyta



(f) Welireg

Figure 2: 3D structures of some drugs

Table 1: Canonical SMILES of selected Drugs

Drugs ID	Canonical Smiles
Sunitinib	CCN(CC)CCNC(=O)C1=C(NC(=C1C)C=C2C3=C(C=CC(=C3)F)NC2=O)C
Pazopanib	CC1=C(C=C(C=C1)NC2=NC=CC(=N2)N(C)C3=CC4=NN(C(=C4C=C3)C)C)S(=O)(=O)N
Cabozantinib	COC1=CC2=C(C=CN=C2C=C1OC)OC3=CC=C(C=C3)NC(=O)C4(CC4)C(=O)NC5=CC=C(C=C5)F
Sorafenib	CNC(=O)C1=NC=CC(=C1)OC2=CC=C(C=C2)NC(=O)NC3=CC(=C(C=C3)Cl)C(F)(F)F
Axitinib	CNC(=O)C1=CC=CC=C1SC2=CC3=C(C=C2)C(=NN3)/C=C/C4=CC=CC=
Lenvatinib	COC1=CC2=NC=CC(=C2C=C1C(=O)N)OC3=CC(=C(C=C3)NC(=O)NC4CC4)Cl
Tivozanib	CC1=CC(=NO1)NC(=O)NC2=C(C=C(C=C2)OC3=C4C=C(C(=CC4=NC=C3)OC)OC)Cl
Temsirolimus	C[C@@H]1CC[C@H]2C[C@@H](/C(=C/C=C/C=C/[C@H](C[C@H](C(=O)[C@@H]([C@@H](/C(=C/[C@H](C(=O)C[C@H](OC(=O)[C@@H]3CCCCN3C(=O)C(=O)[C@@]1(O2)O)[C@H](C)C[C@@H]4CC[C@H]([C@@H](C4)OC)OC(=O)C(C)(CO)CO)C)/C)O)OC)C)C)/C)OC
Everolimus	C[C@@H]1CC[C@H]2C[C@@H](/C(=C/C=C/C=C/[C@H](C[C@H](C(=O)[C@@H]([C@@H](/C(=C/[C@H](C(=O)C[C@H](OC(=O)[C@@H]3CCCCN3C(=O)C(=O)[C@@]1(O2)O)[C@H](C)C[C@@H]4CC[C@H]([C@@H](C4)OC)OCCO)C)/C)O)OC)C)C)/C)OC
Belzutifan	CS(=O)(=O)C1=C2[C@@H]([C@@H]([C@@H](C2=C(C=C1)OC3=CC(=CC(=C3)CN)F)F)F)O
Sirolimus	C[C@@H]1CC[C@H]2C[C@@H](/C(=C/C=C/C=C/[C@H](C[C@H](C(=O)[C@@H]([C@@H](/C(=C/[C@H](C(=O)C[C@H](OC(=O)[C@@H]3CCCCN3C(=O)C(=O)[C@@]1(O2)O)[C@H](C)C[C@@H]4CC[C@H]([C@@H](C4)OC)O)C)/C)O)OC)C)C)/C)OC
Quercetin	C1=CC(=C(C=C1C2=C(C(=O)C3=C(C=C(C=C3O2)O)O)O)O)O
Savolitinib	C[C@@H](C1=CN2C=CN=C2C=C1)N3C4=NC(=CN=C4N=N3)C5=CN(N=C5)C
Gemcitabine	C1=CN(C(=O)N=C1N)[C@H]2C([C@@H]([C@H](O2)CO)O)(F)F
Capecitabine	CCCCCOC(=O)NC1=NC(=O)N(C=C1F)[C@H]2[C@@H]([C@@H]([C@H](O2)C)O)O
5-Fluorouracil	C1=C(C(=O)NC(=O)N1)F
Inlyta	CNC(=O)C1=CC=CC=C1SC2=CC3=C(C=C2)C(=NN3)/C=C/C4=CC=CC=N4
Welireg	CS(=O)(=O)C1=C2[C@@H]([C@@H]([C@@H](C2=C(C=C1)OC3=CC(=CC(=C3)CN)F)F)F)O
Cabometyx	COC1=CC2=C(C=CN=C2C=C1OC)OC3=CC=C(C=C3)NC(=O)C4(CC4)C(=O)NC5=CC=C(C=C5)F

Table 2: Physicochemical property data for the investigated drug molecules

Drugs	D	IR	MV	ST	P	PSA	MR	logP	logD	BCF	KOC	HBA
A1	1.2	1.611	324.1	51.3	44.6	77	112.5	3.15	-0.10	1.00	1.00	6
A2	1.4	1.702	310.4	56.7	47.6	127	120.2	1.98	2.06	15.27	163.19	9
A3	1.4	1.688	359.0	67.5	54.3	99	137.0	4.84	3.25	111.81	601.61	8
A4	1.5	1.625	319.5	51.8	44.8	92	113.1	5.16	4.26	1010.62	4918.38	7
A5	1.4	1.728	284.8	75.2	45.0	96	113.5	4.15	3.63	334.58	2203.52	5
A6	1.5	1.697	290.6	75.2	44.4	116	112.0	3.39	2.93	97.32	892.53	8
A7	1.4	1.68	320.0	59.7	47.9	108	120.9	4.31	3.98	525.96	2530.88	9
A8	1.2	1.554	853.1	53.8	108.3	242	273.2	2.96	4.12	797.29	4155.04	17
A9	1.2	1.548	811.2	51.4	102.2	205	257.7	3.35	4.25	992.08	4858.76	15
A10	1.6	1.61	244.7	62.2	33.6	96	84.9	1.51	1.39	6.75	136.52	5
A11	1.2	1.551	773.5	51.4	97.8	195	246.7	3.54	4.21	939.11	4671.63	14
A12	1.8	1.823	168.0	114.9	29.1	127	73.3	2.08	1.87	14.99	230.01	7
A13	1.6	1.833	222.5	69.6	38.8	92	98.0	0.54	0.49	1.11	28.93	9
A14	1.8	1.652	142.3	65.3	20.6	108	52.1	-0.47	-1.38	1.00	4.15	7
A15	1.5	1.6	240.5	51.3	32.6	121	82.3	0.97	0.39	1.00	28.84	9
A16	1.5	1.523	84.6	46.1	10.2	58	25.9	-0.78	-0.62	1.00	10.37	4
A17	1.4	1.728	284.8	75.2	45.0	96	113.5	4.15	3.63	334.58	2203.52	5
A18	1.6	1.61	244.7	62.2	33.6	96	84.9	1.51	1.39	6.75	136.52	5
A19	1.4	1.688	359.0	67.5	54.3	99	137.0	4.84	3.25	111.81	601.61	8